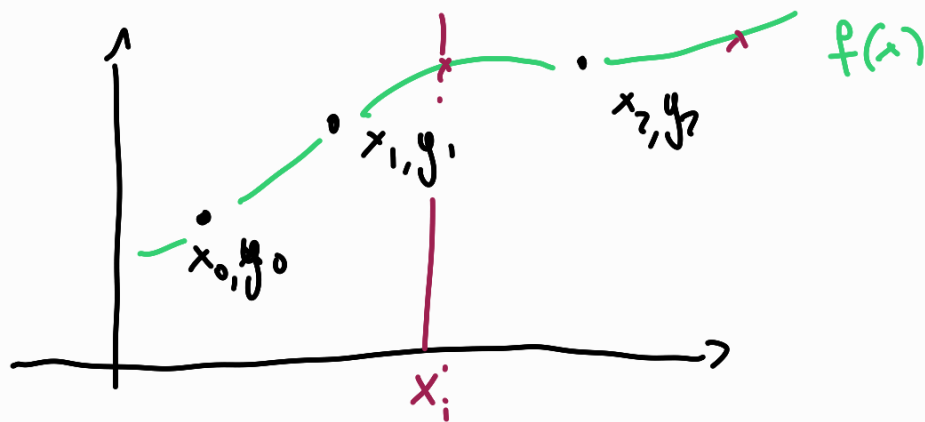


# Approximations of functions and data

$$x_0, x_1, x_2, x_3, \dots, x_n$$

$$y_0, y_1, y_2, \dots, y_n \quad y_i = f(x_i)$$



$$\hat{f}(x) \sim f(x)$$

$\hat{f}$  should be "easy" to evaluate

$\hat{f}$  should be a "good approximation" of  $f$

Interpolation  $\hat{f}(x_i) = f(x_i) = y_i \quad \forall i$

Polynomials

$$\mathbb{P}_q \equiv \left\{ f \in C^\infty(\mathbb{R}) \mid f(x) = \sum_{i=0}^q \alpha_i x^i \right\}$$

## Theorem (Weierstrass)

Given  $f : [a, b] \rightarrow \mathbb{R}$   $f \in C^0([a, b])$

$\forall \varepsilon > 0 \exists p \in \mathbb{N}$  s.t.

$\exists g \in \mathbb{P}_p$  s.t.  $\forall x \in [a, b]$

$$|f(x) - g(x)| < \varepsilon$$

## Lagrange basis

Given  $n+1$  points  $x_0 < x_1 < \dots < x_n$

The Lagrange basis related to the point  $\{x_i\}$  is the set of functions

$$\{l_i\}_{i=0}^n \text{ s.t. } l_i \in \mathbb{P}_n \quad \forall i$$
$$l_i(x_j) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$$l_i(x) = \frac{\prod_{j=0, j \neq i}^n (x - x_j)}{\prod_{j=0, j \neq i}^n (x_i - x_j)}$$

$$l_i(x_j) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases} \quad l_i \in \mathbb{P}_n$$

$\mathbb{P}_n$  is a vector space of dimension  $n+1$

$\{l_i\}_{i=0}^n$  is a set of linearly independent elements of  $\mathbb{P}_{n+1}$

$$\sum_i l_i(x_j) = 1 \quad \text{for all } j$$

$\Rightarrow$  Any element of the space  $\mathbb{P}_n$  can be represented by a linear combination of the basis functions  $l_i$

$$p \in \mathbb{P}_n \Rightarrow \exists \{\alpha_i\} \text{ s.t. } p(x) = \sum_{i=0}^n \alpha_i l_i(x)$$

Let's go back to the interpolation problem

$$\hat{f}(x) \in \mathbb{P}_n \quad \hat{f}(x_i) = y_i = f(x_i)$$

$$\hat{f}(x) = \sum_{i=0}^n \alpha_i l_i(x)$$

what are the values  $\alpha_i$  that enforce interpolation

$$\hat{f}(x_j) = \sum_{i=0}^u \alpha_i \underbrace{l_i(x_j)}_{\delta_{ij}} = \alpha_j$$

$$\boxed{\hat{f}(x_j) = \sum_{i=0}^u y_j l_i(x_j)}$$

- We have proved that the interpolating polynomial of d. u for the data  $(x_i, y_i) \ i=0 \dots u$  does exist
- Proof of uniqueness is trivial

assume  $p \in \mathbb{P}_u$   $p(x_i) = y_i$

$$p(x) - \hat{f}(x) \in \mathbb{P}_u$$

$$p(x_i) - \hat{f}(x_i) = y_i - y_i = 0$$

a polynomial of degree u that is 0 at u+1 points

$$\Rightarrow p(x) - \hat{f}(x) = 0 \quad \forall x$$

Lagrangian interpolation on equispaced points

$$f: [a, b] \longrightarrow \mathbb{R} \quad u \in \mathbb{N} \quad h = \frac{b-a}{u}$$

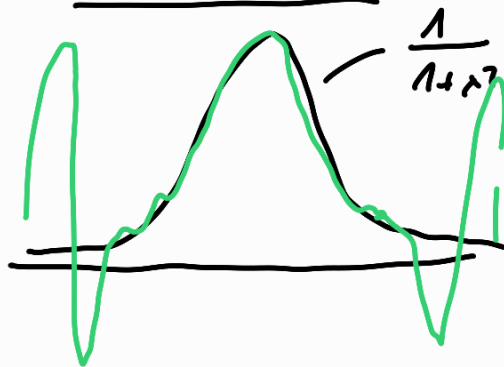
$$a_i = h \cdot i \quad i = 0 \dots u-1$$

$$b_i = h + a_i \quad i = 0 \dots u-1$$

$$x_i = h \cdot i \quad i = 0 \dots u$$

$$\hat{f}(x) \quad \hat{f}(x_i) = f(x_i) \quad \exists! \hat{f} \in \mathbb{P}_u$$

Runge Phenomenon



large errors near  
the ends of the  
interval

Error estimates for Lagrangian interpolation

$$f : [a, b] \longrightarrow \mathbb{R} \quad f \in C^{u+1}([a, b])$$

$$(\pi_u f)(x) \in \mathbb{P}_u \quad \text{interpolating polynomial}$$

of degree  $u$  on  $u+1$  equally spaced nodes  $x_i$

$$(E_u f)(x) = f(x) - (\pi_u f)(x)$$

Proposition  $\forall x \in [a, b] \quad \{x_i\} \in [a, b]$

$$\text{s.t.} \quad (E_n f)(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \prod_{i=0}^n (x - x_i)$$

$$|(E_n f)(x)| \leq \sup_{x \in [a, b]} |f^{(n+1)}(x)| \prod_{i=0}^n \frac{|x - x_i|}{(n+1)!}$$

$$\left| \prod_{i=0}^n (x - x_i) \right| \leq n! \frac{h^{n+1}}{4} \quad h = x_{i+1} - x_i$$

$$\max_{x \in [a, b]} |E_n f(x)| \leq \max_{x \in [a, b]} \left| \frac{f^{(n+1)}(x)}{(n+1)!} \right| n! \frac{h^{n+1}}{4}$$

$$= \max_{x \in [a, b]} \frac{|f^{(n+1)}(x)|}{4(n+1)} \cdot h^{n+1}$$

for some functions the term in red increases

very fast when increasing  $n \Rightarrow$

$$\Rightarrow \max_x |E_n f(x)| \xrightarrow{n \rightarrow \infty} \infty$$